Failure of the Change of Sign Method

Exam code: 9MA0

Download PDF

Written by: Lucy Kirkham
Updated on 16 February 2026

Did this video help you? Yes No

Failure of the change of sign method

When can the change of sign method fail?

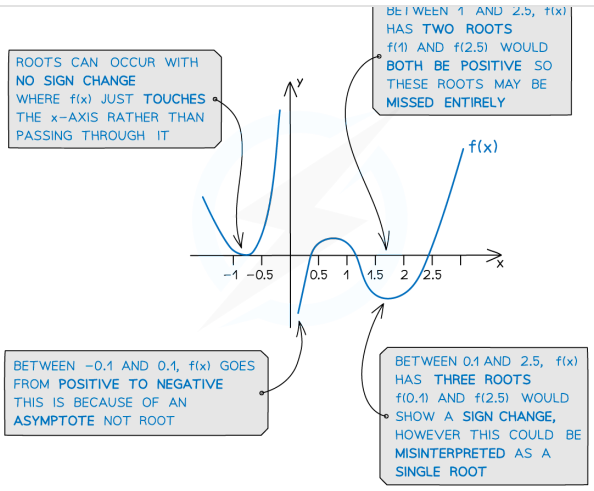
- There are certain circumstances where a **change of sign** fails to appropriately reflect the number or existence of roots
- If the interval is **too large** there may be more than one root within it
  - An **even number of roots** mean roots are missed entirely as a sign change is not identified
  - An **odd number of roots (> 1)** may mean not all roots are identified
- In a discontinuous funciton, a sign change may occur which is not caused by a root but by an asymptote
- If a function **touches the x-axis** but does not pass through it there would be a root but no sign change

Help me understand

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it



Copyright © Save My Exams. All Rights Reserved

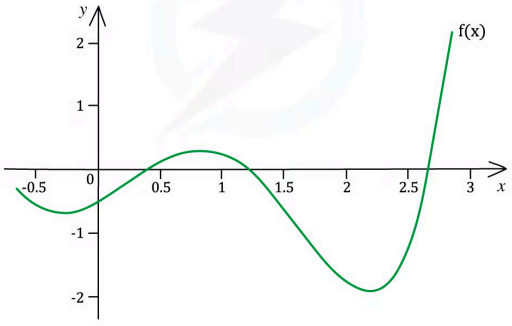
Worked Example

The graph shows the function of $f(x) = xe^x - 2x^3 - 0.5$ for $-0.5 \le x \le 3$.

a) State how many roots the equation $xe^x - 2x^3 - 0.5 = 0$ has in the interval $-0.5 \le x \le 3$.

b) Give a reason why using a sign change method to check for a root over the interval $-0.5 \le x \le 3$ will fail.

c) By using an appropriate interval find the first positive root to 1 decimal place for the function $f(x) = xe^x - 2x^3 - 0.5$.



Copyright © Save My Exams. All Rights Reserved
save my exams

WILL REFLECT 3 ROOTS NOT JUST 1

c) $0.4e^{0.4} - 2(0.4)^3 - 0.5 = -0.031...$

$0.5e^{0.5} - 2(0.5)^3 - 0.5 = 0.074...$

SIGN CHANGE METHOD WORKS BEST FOR SMALL INTERVALS

$0.4 < x < 0.5$

$0.45e^{0.45} - 2(0.45)^3 - 0.5 = 0.23$

$0.4 < x < 0.45$ $x = 0.4$ (1 dp)

SIGN CHANGE BETWEEN 0.4 AND 0.45 MEANS ROOT = 0.4 (1 dp)

Copyright © Save My Exams. All Rights Reserved

save my exams

Teacher only

New! Exclusive Teacher-Only Questions

- Fresh, expert-written questions
- Students can't access them
- Fair accurate tests - built fast

Explore in Test Builder →

Test yourself

Was this revision note helpful? Yes No

Previous Next

Resources

Learning Hub

Scholarship

Join

Past Papers

Solution Banks

Sitemap

Dashboard

Overview

Brand ambassador

Profile

Subscription

Settings

Log out

Company

About us

Exam Specificity

Content Quality

Promotions

Jobs

Terms

Privacy

Cookie Policy

Help and Support

Subjects

Biology

Chemistry

Physics

Maths

Geography

English Literature

Psychology

All Subjects

© Copyright 2015–2026 Save My Exams Ltd. All Rights Reserved.

Neither IBO nor College Board were involved in the production of, and do not endorse, the resources created by Save My Exams.